

Vivek Kumar

☎ +91-7870145339 | ✉ vivek.kumar240043@gmail.com | 🔗 LinkedIn | 🐙 GitHub

Quantitative Economics & Data Science student with hands-on experience translating ambiguous analytical problems into structured, data-driven solutions. Skilled in predictive modeling, statistical inference, causal reasoning, and deploying reproducible ML pipelines on real-world datasets across healthcare and commercial domains.

EDUCATION

● **Birla Institute of Technology, Mesra** 2022 – 2027
Integrated M.Sc in Quantitative Economics and Data Science CGPA: 7.05

RESEARCH & PUBLICATIONS

- **Vivek Kumar**, Dr. Mrinal Jana. “Distribution-free Uncertainty Quantification for Option Pricing Using Residual-space Conformal Prediction.” *CIMA 2026*, **Accepted**. Proceedings to appear in Springer *LNNS* (Paper ID-498, Scopus-indexed).
Developed a novel, distribution-free framework for quantifying prediction uncertainty — directly applicable to risk quantification in clinical trial outcomes and healthcare cost forecasting where parametric assumptions often fail.

APPLIED DATA SCIENCE & ANALYTICS PROJECTS

- **End-to-End Decision Science & Analytics Platform** *Python, SQL, Streamlit, GenAI, NetworkX*
 - Translated an ambiguous, unstructured business hypothesis into a structured analytical framework with clearly defined success metrics, data requirements, and evaluation criteria.
 - Designed and deployed a fully automated ETL pipeline ingesting data from 5+ heterogeneous live sources daily, processing 126+ entities with robust data-quality checks and error handling.
 - Developed a multi-factor predictive scoring model combining graph-based features, statistical signals, and NLP-extracted variables, validated via walk-forward backtesting achieving 66.7% out-of-sample accuracy.
 - Built an interactive Streamlit dashboard with Gemini GenAI integration enabling natural language querying of the underlying database for non-technical stakeholders.
 - Operationalized the solution with Docker, CI/CD (GitHub Actions), and AWS EC2 deployment for reproducible, automated daily execution.
- **Bayesian Causal Inference for Epidemiological Modeling** *R, Probabilistic Programming, Healthcare Data*
 - Applied Metropolis-Hastings MCMC to perform Bayesian inference on COVID-19 transmission rates, estimating critical epidemiological parameters (R_0 , recovery rate) from real-world **healthcare datasets**.
 - Designed a causal framework to quantify the effect of non-pharmaceutical interventions on disease spread, moving beyond correlational analysis to inform **data-driven public health decisions**.
 - Evaluated model convergence and posterior uncertainty using trace diagnostics and sensitivity analysis; documented all assumptions, limitations, and data-quality considerations.
- **Long-Form Memory System for LLMs | Neurohack, IIT Guwahati (26th Rank)** *FastAPI, SQLite, NLP, Embeddings*
 - Secured 26th rank nationally (Certificate) by architecting a scalable memory system for LLMs handling 1,000+ conversation turns without performance degradation.
 - Engineered a hybrid retrieval pipeline combining semantic embeddings (cosine similarity) with lexical BM25 ranking via SQLite FTS5, achieving sub-100ms retrieval latency — applicable to GenAI-powered analytics assistants.
- **Optimization via Genetic Algorithms & Scenario Analysis** *Python, SciPy, Pandas, NumPy*
 - Engineered a Real-coded Genetic Algorithm from scratch to solve high-dimensional, non-convex optimization problems — a technique directly applicable to resource allocation and scenario analysis.
 - Conducted rigorous out-of-sample validation using time-series cross-validation, demonstrating algorithmic superiority over baseline methods with quantified performance metrics.

TECHNICAL SKILLS

Languages: Python, R, SQL, C++
Data Science & ML: Predictive Modeling, NLP/GenAI, Segmentation, Feature Engineering, Scikit-learn, XGBoost, TensorFlow, Statsmodels
Statistics & Decision Science: Causal Inference, Bayesian Methods, A/B Testing, Hypothesis Testing, Experiment Design, Regression, Time Series Forecasting
Data Engineering & MLOps: Pandas, NumPy, SQLite, ETL Pipelines, Web Scraping, API Integration, Git, Docker, CI/CD (GitHub Actions), AWS EC2
Visualization & Communication: Streamlit, Plotly, Matplotlib, Seaborn, Stakeholder Presentations
Relevant Coursework: Statistical Inference, Econometrics, Probability Theory, Time Series Analysis, Machine Learning, Multivariate Data Analysis, Optimization, Algorithms for Big Data

●Events & Public Relations Organizer | *Data Science Summit '23, BITOTSAV '24*

- Spearheaded corporate outreach for the Data Science Summit (DSS 23), successfully pitching sponsors and coordinating cross-functional teams — demonstrating clear communication with non-technical stakeholders.
- Organized logistics and event management for Pantheon 2023 and BITOTSAV 2024 college festivals.